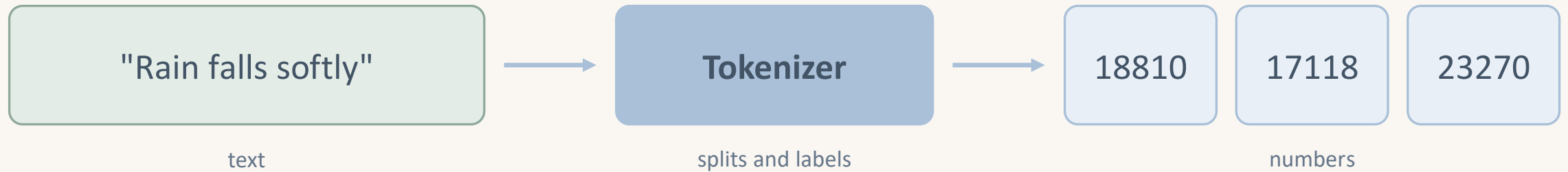


Tokenizers: How Language Becomes Numbers

The first step in teaching a computer to read



What is a tokenizer?

1

Computers can't read words.
They only work with numbers.

2

A tokenizer cuts text into tokens.
A token is a small piece of text.

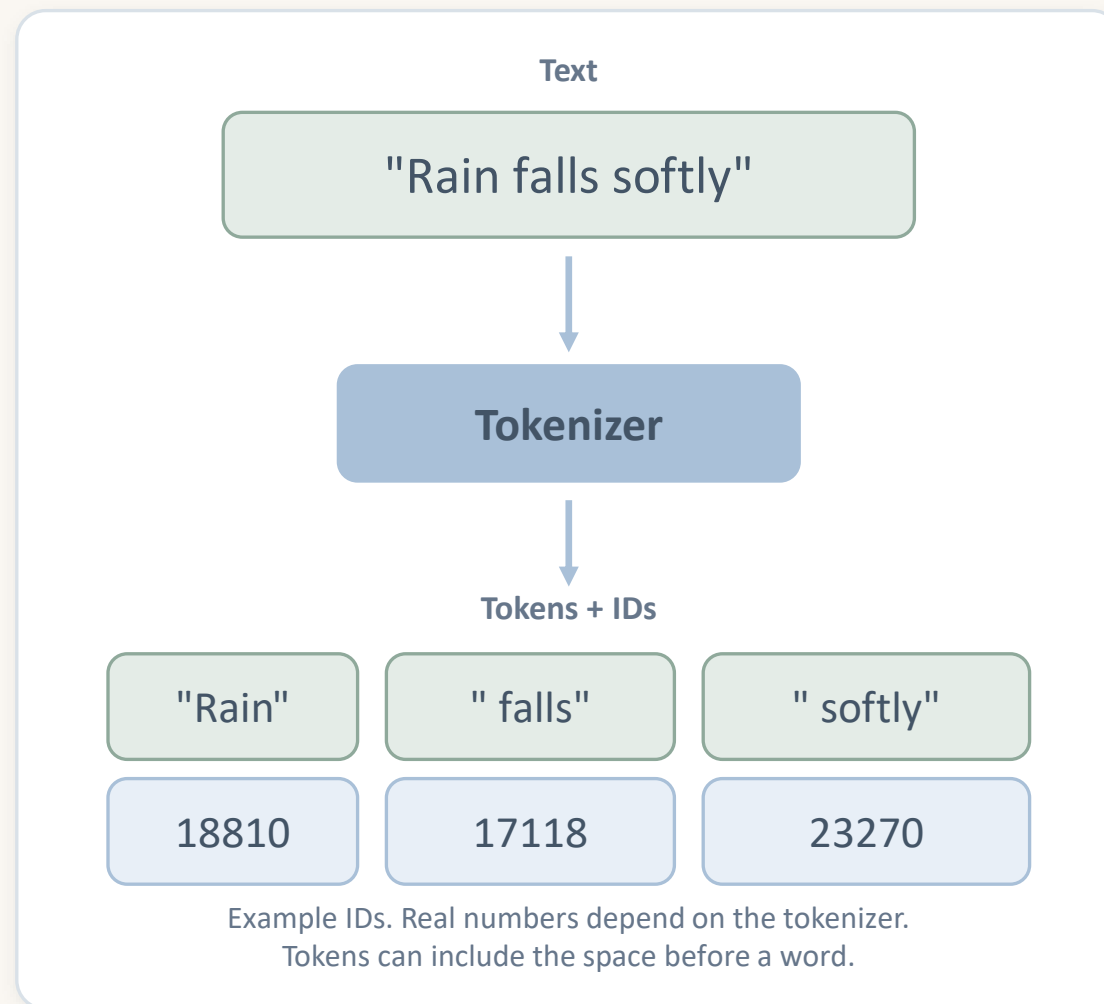
3

Each token gets an ID.
An ID is a number, like a name tag.

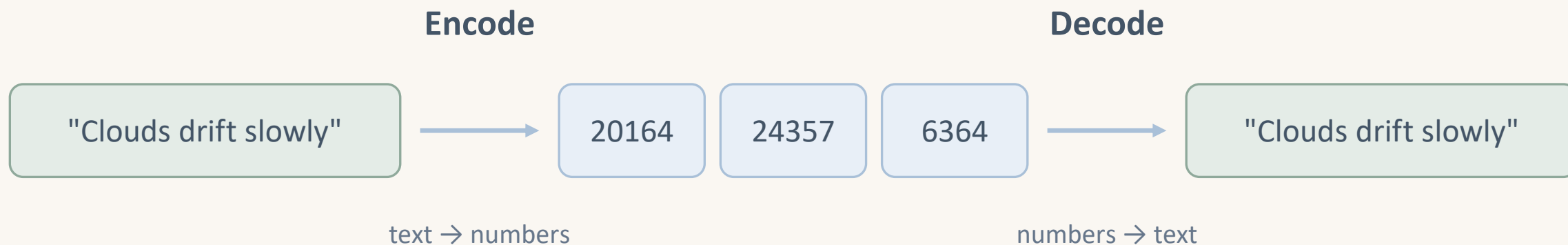
KEY IDEA



A tokenizer translates text into numbers.



Encoding and decoding



KEY IDEA

The model works with numbers, not words.

Different kinds of tokenizers

Each kind splits the same word differently:

"sunlight"

Character tokenizer

Every letter is its own token.

s u n l i g h t

8 tokens

+ Simple

- Makes many tokens

Word tokenizer

Every whole word is one token.

sunlight

1 token

+ Easy to understand

- Stuck on new words

Byte Pair Encoding (BPE) tokenizer

Common word pieces are tokens. It learns them from lots of text.

sun light

2 tokens

+ Handles new words

+ Keeps token count low

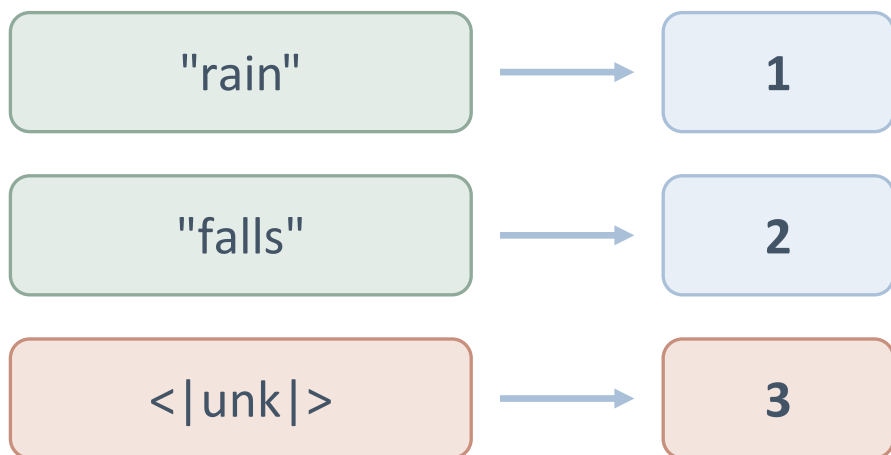
Note: exact BPE pieces depend on the tokenizer. GPT models use BPE.

Other subword tokenizers work in a similar way, such as WordPiece (used by BERT) and Unigram (used by T5).

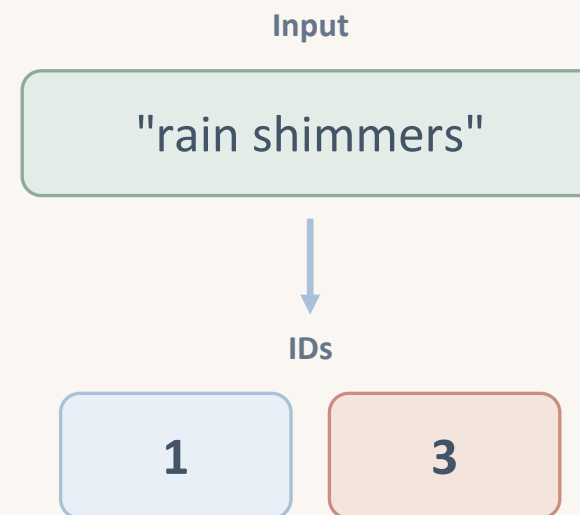
Vocabulary and unknown tokens

A vocabulary is a lookup list. Each text piece has its own ID.

Vocabulary



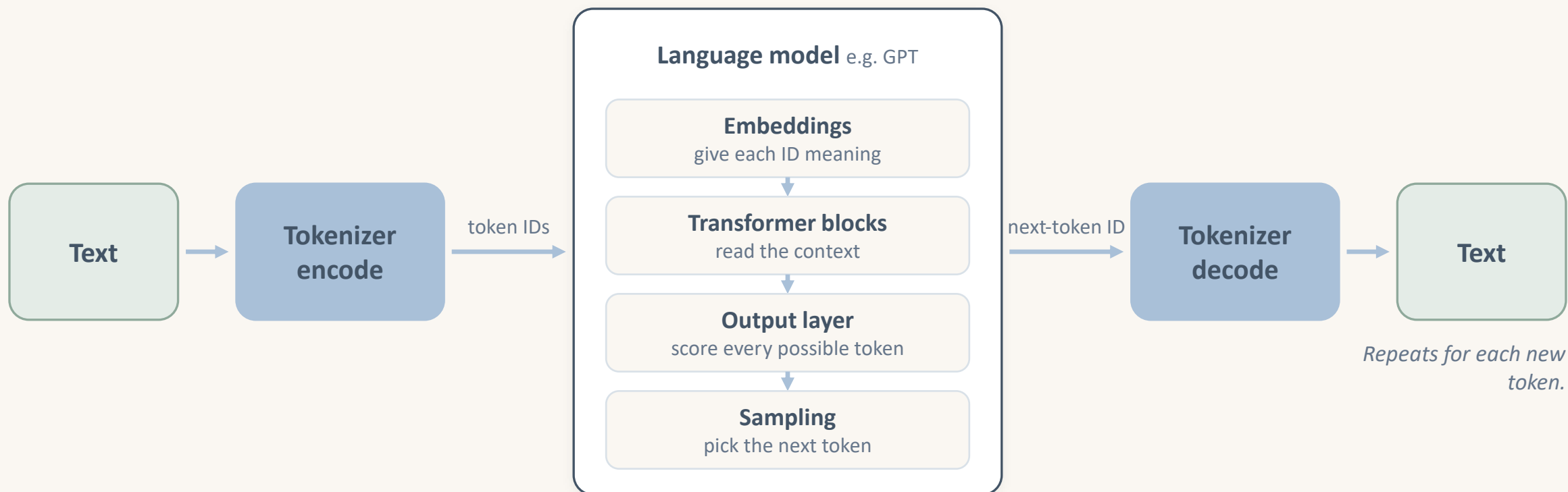
<|unk|> is a special token for unknown words.



"shimmers" is not in the vocabulary, so it becomes <|unk|>.

Why tokenizers matter for LLMs

LLM = large language model. The tokenizer is one part: it works at the start and the end.



TAKEAWAY

Before an LLM can generate language, it must first turn language into tokens.